SPECIAL VARIETIES

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Contents

1. Fano varieties

Definition 1.0.1. (♠ TODO: projective variety) (♠ TODO: what is the actual definition of fano variety) (♠ TODO: ample divisor)

A *Fano variety over a field k* is a projective variety X over k such that the anticanonical class (Definition 1.0.2) $-K_X$ is an ample divisor.

The variety X is a *Fano variety* if X is smooth, or more generally has mild singularities, and its anticanonical class $-K_X$ is an ample divisor.

Definition 1.0.2. (♠ TODO: variety) Let X be a variety over a field. The *canonical class K_X* is the divisor class associated to the sheaf of terminal n-forms $\omega_X = \bigwedge^n \Omega^1_{X/k}$, where $n = \dim X$. The *anticanonical class* is defined as $-K_X$.

Definition 1.0.3. (♠ TODO: variety) (♠ TODO: cartier divisor, ample, picard group) Let X be a normal Fano variety (Definition 1.0.1) over an algebraically closed field.

The *index of X* is the largest integer r such that $-K_X \sim rH$ for some ample Cartier divisor H in the Picard group $\mathrm{Pic}(X)$.

Proposition 1.0.4. Let X be a smooth Fano variety (Definition 1.0.1) over the complex numbers $\mathbb{C}$. Then $X(\mathbb{C})$ is simply connected and the sheaf cohomology groups satisfy $H^i(X, \mathcal{O}_X) = 0$ for all $i > 0$.

Theorem 1.0.5. (♠ TODO: cite campana 1992, kollar-miyaoka-mori 1992) Let X be a smooth Fano variety (Definition 1.0.1) over an algebraically closed field of characteristic 0. Then X is (♠ TODO: rationally connected).

Appendix A. Miscellaneous definitions